

Name of the Course : **B.Sc. (Hons.) Mathematics CBCS**
Semester : **IV**
Unique Paper Code : **32351402_OC**
Name of the Paper : **C9 - Riemann Integration and Series of Functions**
Duration: **3 Hours** Maximum Marks: **75**

Attempt any four questions. All questions carry equal marks. All symbols have usual meaning.

1. Calculate the upper Darboux sum and lower Darboux sum of the function

$$f(x) = \frac{1}{x^2} \quad \text{and} \quad g(x) = xe^x \quad \text{on the interval } [1, 2] \text{ for a partition}$$

$$P = \{1, 1.25, 1.5, 1.75, 2\}.$$

Calculate upper and lower Darboux integral of $g(x) = x^2 + 5$ on $[2, 4]$. Is g integrable ?

- Let $f: [a, b] \rightarrow R$ be a bounded function. Suppose that there is a partition P of $[a, b]$ such that

$$L(f, P) = U(f, P).$$

Show that f is a constant function.

2. Define $f(x) = x[x]$ on $[0, 4]$. Show that f is integrable, and evaluate $\int_0^4 f(x)dx$. Give an example where the functions f and g are not Riemann integrable, but $f \cdot g$ is integrable. Let f be a continuous function on R and define

$$F(x) = \int_0^{x^4} f(t)dt \quad \text{for } x \in R$$

show that F is differentiable on R and compute F' .

3. Examine the convergence of following improper integrals

$$\int_0^{\infty} e^{-x} (3x + 2)dx \quad \text{and} \quad \int_0^{\infty} \frac{dx}{\sqrt{3x^4 + 5x}}$$

Using the properties of Gamma integral find the value of

$$\int_0^{\infty} x^5 e^{-4x^2} dx$$

4. Let (f_n) be defined by $f_n(x) = 1 - |1 - x^2|^n \forall x \in [-\sqrt{2}, \sqrt{2}]$

Find the pointwise limit of (f_n) on $[-\sqrt{2}, \sqrt{2}]$.

Does the sequence converge uniformly on this interval? Justify your answer. Show that the sequence (f_n) where $f_n(x) = nxe^{-nx^2}$, $x \geq 0$ is not uniformly convergent on $[0, 2]$.

Show that the sequence $\left\{ \frac{\sin(n^2 x^2 + 1)}{n(n+1)} \right\}$ converges uniformly on R .

5. Examine the convergence of the series of functions $\sum f_n$ where $f_n(x) = \frac{1}{1+x^n}$ and show that convergence is non uniform in $(1, \infty)$ and is uniform in $[a, \infty]$, $a > 1$

• Show that $\sum_{n=1}^{\infty} \frac{\cos nx}{n^3}$ converges uniformly on $\mathbb{R}$ to a continuous function.

• Evaluate the integral $\int_0^1 \sum_{n=1}^{\infty} \frac{x}{(n+x^2)^2} dx$

6. Find the radius of convergence and exact interval of convergence for the following power series

$$\sum_0^{\infty} \frac{4^n}{n^{5n+2}} (x-2)^n \quad \text{and} \quad \sum_0^{\infty} \left[\frac{3+5(-1)^n}{7} \right]^n x^n$$

Write the power series expansion for the integral of the following function:

$$f(x) = \frac{x^2}{3-x^3} \quad ,$$

given that $\frac{1}{1-x} = \sum_0^{\infty} x^n \quad , x \in]-1,1[.$